

Generative AI – The art of AI based creation and innovation

About Instructor:

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About Institute:

NexSkill is Pakistan's leading IT training institute, providing the best IT training in Pakistan. NexSkill offers a variety of professional learning platforms to its students. <https://www.nexskill.com/>

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Orientation - Students - Intro, Education, LinkedIn Profile, Current Skillset?

Questions:

- ✓ Which Use cases – you want to address/solve?
- ✓ Which type of data sources you have?
- ✓ List, all of the Softwares – you use?
- ✓ Which business vertical industry you want to target?
- ✓ What are your Course expectations?

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Orientation - Agenda:

What is AI?

Artificial intelligence (AI) is a field of computer science focused on creating systems that can perform tasks normally requiring human intelligence. These tasks include learning from data, recognizing patterns, making decisions, and understanding language. By mimicking aspects of human thinking, AI enables technologies like smart assistants, recommendation systems, autonomous vehicles, and more, helping machines adapt and improve as they process new information.

<https://zapier.com/blog/what-is-ai/>

<https://www.ibm.com/think/topics/artificial-intelligence>

<https://cloud.google.com/learn/what-is-artificial-intelligence?hl=en>

<https://www.mtu.edu/computing/ai/>

<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-ai>

What is Gen-AI?

Generative AI is a branch of artificial intelligence that focuses on creating new content—such as text, images, audio, code, or video—by learning patterns from large datasets. Instead of simply analyzing or recognizing existing information, generative AI models can produce original outputs that resemble human-created work. Technologies like GPT, image generators, and music-creation models are examples of generative AI, enabling applications such as creative writing assistance, realistic image generation, and automated content creation.

<https://zapier.com/blog/generative-ai/>

<https://www.ibm.com/think/topics/generative-ai>

<https://research.ibm.com/blog/what-is-generative-AI>

<https://cloud.google.com/use-cases/generative-ai?hl=en>

<https://aws.amazon.com/what-is/generative-ai/>

Important Use case of Gen-AI?

A key use case of generative AI is **content creation**, where the technology helps produce high-quality text, images, videos, or code with minimal human effort. This allows businesses to automate tasks such as drafting emails, generating marketing materials, designing visuals, or even creating software prototypes. By speeding up creative workflows and reducing manual effort, generative AI enables teams to focus more on strategy and innovation while maintaining consistency and efficiency in their output.

<https://www.ibm.com/think/topics/generative-ai-use-cases>

<https://cloud.google.com/transform/101-real-world-generative-ai-use-cases-from-industry-leaders>

<https://aws.amazon.com/ai/generative-ai/use-cases/>

<https://www.moveworks.com/us/en/resources/blog/transformational-generative-ai-use-cases-examples>

<https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>

<https://www.sap.com/mena/products/artificial-intelligence/what-is-generative-ai.html>

The – Search Engine vs LLM – The Future?

The future of information access is increasingly shaped by the relationship between traditional search engines and large language models (LLMs). Search engines excel at retrieving links and facts from across the web, while LLMs generate natural, conversational responses by understanding context and synthesizing information. Rather than replacing search, LLMs are evolving into intelligent intermediaries that refine queries, summarize results, and deliver clearer insights. As both technologies continue to converge, the future likely lies in hybrid systems where search engines provide updated, reliable data and LLMs transform it into personalized, human-like answers—making information discovery faster, smarter, and more intuitive.

References:

<https://alluredigital.net/blog/search-engines-vs-large-language-models/>

<https://medium.com/@vibhasharma2296/llms-v-s-search-engines-google-v-s-chatgpt-722253f34885>

<https://www.imd.org/ibymd/artificial-intelligence/battle-for-search-dominance/>

<https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/new-front-door-to-the-internet-winning-in-the-age-of-ai-search>

What is LLM?

A large language model (LLM) is an advanced type of artificial intelligence designed to understand, generate, and interpret human language. Trained on vast amounts of text data, LLMs can perform tasks such as answering questions, summarizing information, writing content, translating languages, and engaging in conversation with human-like fluency. By recognizing patterns in language and predicting the most likely words or ideas that follow, LLMs enable more natural and intelligent interactions between humans and computers.

Wiki Web site

https://en.wikipedia.org/wiki/Large_language_model

References - What is LLM?

<https://www.servicenow.com/ai/what-is-llm.html>

<https://www.ibm.com/think/topics/large-language-models>

<https://www.cloudflare.com/learning/ai/what-is-large-language-model/>

<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-are-large-language-models-llms>

<https://www.geeksforgeeks.org/artificial-intelligence/large-language-model-llm/>

Important Platform and services of Gen-AI?

MidJourney

MidJourney.com is the official platform for **MidJourney**, a generative AI system that creates high-quality images from simple text prompts. It is developed by an independent research lab and is known for producing artistic, stylized, and visually rich outputs. MidJourney is primarily accessed through Discord, where users type prompts to generate images, though a dedicated web interface is also emerging. The tool is popular among designers, artists, and creators for its ability to transform imagination into detailed visuals, making it one of the leading AI image generators today.

Official web site:

<https://www.midjourney.com/>

Wiki Web site:

<https://en.wikipedia.org/wiki/Midjourney>

References - What is MidJourney?

<https://www.datacamp.com/tutorial/how-to-use-midjourney-a-comprehensive-guide-to-ai-generated-artwork-creation>

<https://zapier.com/blog/how-to-use-midjourney/>

<https://www.geeksforgeeks.org/artificial-intelligence/what-is-midjourney-ai-and-how-does-it-work/>

<https://www.danthree.studio/en/blog-cgi/what-is-ai-midjourney-and-how-does-it-work>

<https://integrately.com/blog/what-is-midjourney-ai>

DeepSeek

DeepSeek is a Chinese artificial intelligence company based in Hangzhou that develops highly efficient large language models (LLMs) designed to rival leading Western AI systems. Founded in 2023 by Liang Wenfeng, the company gained global attention after releasing its DeepSeek-R1 model and chatbot in January 2025, offering performance comparable to models like GPT-4 and o1 but at a dramatically lower training cost. DeepSeek is known for its open-weight approach, cost-effective training techniques—such as using Mixture-of-Experts architectures—and its focus on transparency, efficiency, and broad accessibility. Its rapid rise disrupted the AI industry, even triggering major shifts in global tech markets due to its ability to deliver high-quality AI at a fraction of traditional resource requirements.

Official Web site:

<https://www.deepseek.com/en/>

Wiki Web Site:

<https://en.wikipedia.org/wiki/DeepSeek>

References - What is DeepSeek?

<https://zapier.com/blog/what-is-deepseek/>

<https://www.techtarget.com/whatis/feature/DeepSeek-explained-Everything-you-need-to-know>

<https://mindflow.io/blog/deepseek-vs-openai-what-is-deepseek-what-does-deepseek-do>

<https://kalm.works/en/contents/technology/what-is-deepseek-differences-from-chatgpt-and-use-cases>

<https://www.bakerbotts.com/thought-leadership/publications/2025/january/what-is-deepseek-and-why-does-it-matter>

OpenAI

OpenAI is an American artificial intelligence research organization based in San Francisco, focused on developing safe and beneficial artificial general intelligence (AGI). Founded in 2015 by leaders in technology and research, the company is known for creating influential AI systems such as the GPT family of language models, the DALL-E image generators, and the Sora text-to-video model. Its mission centers on ensuring that advanced AI benefits all of humanity, and it operates through a hybrid structure that includes both a nonprofit foundation and a controlled for-profit public benefit corporation. OpenAI's work has played a major role in popularizing generative AI worldwide—especially after the launch of ChatGPT in 2022, which sparked global interest in conversational AI and accelerated industry adoption of large-scale AI models.

Official Web Site:

<https://openai.com>

Wiki Web site:

<https://en.wikipedia.org/wiki/OpenAI>

References - What is OpenAI?

<https://www.moveworks.com/us/en/resources/ai-terms-glossary/openai>

<https://www.coursera.org/articles/what-is-openai>

<https://kaopiz.com/en/articles/what-is-openai/>

<https://www.infobip.com/glossary/open-ai>

What is SaaS? What is Subscription Model?

Software as a Service (SaaS) is a cloud-based software delivery model in which applications are hosted online and accessed by users through a web browser rather than installed on local computers. This approach allows organizations and individuals to use powerful software without managing hardware, maintenance, or updates—everything is handled by the service provider. SaaS offers flexibility, scalability, and cost-effectiveness, making it popular for tools like email services, CRM systems, collaboration platforms, and business applications across many industries.

<https://www.salesforce.com/ap/saas/>

<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-saas>

<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-saas>

<https://aws.amazon.com/what-is/saas/>

<https://www.sap.com/resources/what-is-saas>

<https://www.oracle.com/africa/applications/what-is-saas/>

What is Gen-AI based “Web Development”?

Generative AI-based web development refers to the use of AI models that can automatically generate code, layouts, content, and design elements for websites based on natural-language instructions. Instead of manually writing HTML, CSS, JavaScript, or backend logic, developers can describe what they want, and the AI produces functional components, UI designs, and even full web pages. This approach speeds up development, reduces repetitive tasks, and empowers both technical and non-technical users to build modern, responsive web experiences with greater efficiency and creativity.

<https://www.digitalocean.com/resources/articles/ai-website-builders>

<https://zapier.com/blog/best-ai-website-builder/>

Every tool – merging - Gen-AI vs no-code vs low-Code vs detail code

The Bubble.io way:

<https://github.com/ShahzadSarwar10/LOW-CODE-NO-CODE-B-1/tree/main/Bubble/Week0>

The Webflow way:

<https://github.com/ShahzadSarwar10/LOW-CODE-NO-CODE-B-1/tree/main/Webflow/Week0>